Q1 Presentation Script

Feeding America's need estimates vs. The Food Group's food shelf activity · 9 slides + live map

How to use this script

This is the full, long version of what to say for each Q1 slide, with the reasoning behind every claim, so you can study it and cut it down yourself. Each slide has four parts:

- **Say this** (blue box): the spoken script. **[Bold orange brackets]** are actions, not words to say.
- **What it means:** the idea behind the slide in plain words, so you can explain it even if you forget the script.
- **Numbers to know:** the figures on the slide and where they come from.
- **If a judge asks:** short answers to the follow-up questions that slide is most likely to get.

Files

What	File
Slides	outputs/q1/Q1_slides.pptx
Live map	outputs/q1/q1_interactive_maps.html
Full analysis	notebooks/q1_need_vs_foodshelf_activity.ipynb
Study guide	docs/Q1_study_guide.pdf

Before you start

1. Open the map in Chrome and press F11 for full screen.
2. Leave it on tab 1, **Which counties?**, with the slider at 2×.
3. Open the slides in PowerPoint and start the slideshow.
4. Switch between the two with **Alt+Tab**. On the map, keys **1, 2, 3** jump between tabs.
5. Three slides are live demos: slide 4 (tab 1), slide 6 (tab 2), slide 7 (tab 3). Each has a LIVE MAP tag. If the map fails, stay on the slide: it shows the same result as a picture.

Words to use consistently

- **Need** = Feeding America's Map the Meal Gap estimate of food-insecure people. A model estimate, not a count.
- **Activity** = what food shelves recorded: visits (household trips), pounds of food, individuals served.
- **Expected** = what counties with a similar number of food-insecure people typically record.
- **Lower / higher than expected** = a county records less or more than that. It doesn't automatically mean people are under-served or over-served.

Slide 1: Does estimated need match food shelf activity?

On the slide: the question, the two data sources and the years.

SAY THIS

Question 1 asks whether two sets of numbers tell the same story. The first is Feeding America's Map the Meal Gap, which estimates how many people in each county are food insecure. The second is The Food Group's own food shelf data: how many visits, pounds of food and individuals each food shelf recorded.

We compared them for all 87 Minnesota counties, for 2022, 2023 and 2024. Those are the years where we have both.

We asked four things: do they tell the same story, which food shelf metric matches need best, which counties break the pattern, and why.

What it means

The two sources measure different things. Map the Meal Gap is a **model estimate** of people. The food shelf data counts **trips, pounds and people served**, including repeat visits. Because the units differ, we never subtract one from the other. "Same story" means they **move together**: the same counties rank high on both, and both rise or fall together.

Numbers to know

- 87 counties × 3 years = 261 county-years; 258 compared, because Wilkin has no food shelf in the data.
- Map the Meal Gap 2025 release covers 2022–2023; the 2026 release covers 2024.

If a judge asks

Why only 2022–2024?

That's the only overlap. Map the Meal Gap has county estimates for 2022–2024; the food shelf data runs to mid-2026, but there are no need estimates for 2025–2026 yet.

Did you clean the data?

Yes. Typos were removed (for example, a Waseca month reporting 273,385 households), duplicates were merged, and blank months were kept as missing, not zero. Removing the 7 county-years with incomplete reporting changed no correlation by more than 0.03.

Slide 2: Same story? Only partly

On the slide: a line chart of statewide totals (2022 = 100) and three verdicts: county size □, direction over time □, per resident □.

SAY THIS

The short answer is: only partly. We looked at it three ways.

First, by county size. Counties with more food-insecure people do have more visits, more pounds and more individuals served. The correlation is about 0.9, which sounds very strong. But when we replaced the need estimate with plain population, which knows nothing about food insecurity, it predicted food shelf activity just as well. So most of that agreement is simply that big counties have more of everything.

Second, over time. Both went up from 2022 to 2024. Estimated need rose 25%, while visits rose 46%, pounds 31% and individuals 60%. So the direction agrees, but food shelf activity grew faster than the estimate.

Third, per resident, which is the fair comparison once you remove county size. Here there's no relationship. Counties with a higher food insecurity rate don't have more food shelf use per person. By 2024 the correlation is essentially zero.

What it means

A correlation tells you whether two things move together, from −1 to +1, where 0 means no relationship. We used **Spearman's correlation**, which compares rankings. That matters in Minnesota, because Hennepin is so large it would dominate a normal correlation.

The key move on this slide is the **population benchmark**. A high correlation on totals looks like agreement, but if population alone gets the same number, the need estimate isn't adding anything. The honest test is **per resident**: activity per 1,000 residents against the food insecurity rate. That's where the two stories split.

"Individuals" grows fastest partly because it counts repeat visits: someone who comes every month counts 12 times a year.

Numbers to know

Measure	Result
Totals correlation (Spearman)	0.89 to 0.92
Same, using population only	0.88 to 0.92
Need estimate vs. population (ranking)	0.99
Per resident, visits 2022 / 2023 / 2024	0.12 / 0.08 / −0.01
Change 2022→2024: need / visits / pounds / individuals	+25% / +46% / +31% / +60%

If a judge asks

Why compare per resident at all?

Because totals mostly measure how big a county is. Per resident removes size and asks the real question: do needier places use food shelves more?

Does this mean the Feeding America estimate is wrong?

No. It means county need estimates alone don't predict where food shelf activity happens. Other things, like where food shelves are, matter too. That's what slides 5 to 8 show.

Do the three metrics agree with each other?

Yes. Per resident they correlate 0.79 to 0.96, so all three tell one story, and it's that shared story that differs from the need estimate.

Slide 3: Pounds tracks need best, by a small margin

On the slide: a bar chart of correlations for visits, pounds and individuals, on totals and per resident, plus cards "+0.02" and "0.9 → 0.1".

SAY THIS

The second part of the question asks which food shelf metric lines up most and least with Feeding America's estimates.

Pounds distributed is consistently the strongest, and individuals served is usually the weakest, with visits close behind. Pounds came out on top in every year and with both types of correlation we used.

But the lead is small. On totals, pounds is ahead by only about 0.02. When we reshuffled the counties 2,000 times to test how stable that is, pounds usually stayed ahead, but the ranges overlap. So we'd describe it as consistently strongest, but not by much.

The bigger lesson is on the right of the chart. On totals, all three metrics correlate around 0.9. Per resident, all three drop to about 0.1 or less. So whether you adjust for county size matters far more than which metric you pick.

What it means

We checked the ranking four ways so it wouldn't rest on luck:

1. **Bootstrap:** re-drew the 86 counties at random 2,000 times. Pounds beat the others in 83–97% of draws on totals, but per resident every difference could be zero.
2. **Leave-one-out:** dropped each county in turn. Hennepin inflates the totals; Mahnomen drives the one significant per-resident result (pounds 2022 falls from 0.32 to 0.12 without it).
3. **Incomplete reporting:** removed the 7 county-years with missing months. Nothing changed by more than 0.03.
4. **Two correlation types:** Spearman and Pearson rank the metrics the same way.

A possible reason pounds does best: pounds reflects how much food the network sends into a county, which is planned roughly around population and need. Visits and individuals depend on how often people return and how each site counts them. This is an interpretation, not something we tested.

Numbers to know

Correlation with need (avg. 2022–2024)	Visits	Pounds	Individuals
Totals	0.89	0.92	0.90
Per resident	0.06	0.10	0.05

If a judge asks

So should The Food Group track pounds?

Pounds is slightly closer to need, but none of the three is a good guide per resident. The bigger lesson is not to plan from county need estimates alone.

Is 0.92 vs 0.90 meaningful?

Not strongly. It's consistent but small, which is exactly how we describe it.

What does "Visits" mean here?

The brief defines it as household trips. The Food Group's published visit counts match individuals instead. Our conclusions hold under both definitions.

Slide 4: 12 counties break the pattern LIVE MAP

On the slide: a bar chart of the 12 counties (orange = records less than expected, blue = more) and two cards with the rules. The map's tab 1 shows the same counties and lets you move the cutoff.

SAY THIS

The third part asks where the relationship breaks down.

[Alt+Tab to the map. Tab 1, Which counties?, is already open at 2×.]

We can't compare raw totals, because big counties always record more. So for each county we asked: what do counties with a similar number of food-insecure people usually record? That's the county's expected activity. Then we compared what it actually recorded with that.

We flagged a county when it recorded at least twice, or half or less, of what's expected. To make the shortlist, it had to do that on at least two of the three metrics, in at least two of the three years, and agree under a second calculation method.

12 counties pass. Five record less than expected: Nicollet, Pope, Lac qui Parle, Norman and Roseau. Seven record more: Steele, Washington, Rice, Renville, Mahnomen, Pipestone and Olmsted. Steele is the most extreme, at about four times what similar counties record.

[Drag the slider to 3×, then back to 2×.]

The two-times cutoff is a judgment call, so we tested it. At three times, Steele, Washington and Rice are still there. At a looser one and a half times, the list grows, but the same core counties stay. So the finding doesn't hang on the exact cutoff.

[Alt+Tab back to the slides.]

What it means

"Expected" comes from a line fitted through all counties: activity against the number of food-insecure people, on a log scale, for each year and metric. A county's gap is how far it sits above or below that line. A worked example from 2024 visits:

County	Food-insecure people	Expected visits	Actual visits	Ratio
Nicollet	~3,840	~9,600	2,833	0.30×
Steele	~4,100	~10,400	59,192	5.7×

Same size of need, about 20 times different activity. That's what a mismatch is. The ratios on the slide (Nicollet 0.29×, Steele 4.13×) are averages over all three metrics and years, so they differ slightly from a single year.

"Lower than expected" means the county's food shelves **record** less. It doesn't automatically mean people there are under-served; they may be served next door (slide 6).

Numbers to know

Cutoff	Counties shortlisted
1.5×	28
1.75×	17

Cutoff	Counties shortlisted
2× (ours)	12
2.5×	6
3×	3 (Steele, Washington, Rice)

- 26 counties are persistent on at least one metric: 10 lower, 16 higher. None ever switches direction.
- Every shortlisted county reported all 12 months in every year, with at most 8% of site-months missing.

If a judge asks

Isn't 2× arbitrary?
It's a judgment call, but a transparent one: one doubling step. The slider shows the core counties stay at any cutoff, and the explanations on slide 5 use the continuous gaps, not the cutoff.

Couldn't it be bad data?
We checked. No shortlisted county has missing reports. Three counties whose low numbers line up with missing months (Douglas, Grant, Stevens) are flagged as possible data-quality cases and not interpreted.

Why not just divide activity by need?
We tried that, and a per-resident version. Both are biased by size: they call big counties "high" and small ones "low". Our line-based method is size-neutral, and all three methods rank counties similarly (0.86–0.92).

Slide 5: Access explains the gaps. Hardship doesn't

On the slide: a bar chart of how strongly each county characteristic lines up with the mismatch (dark = significant, grey = not), plus cards "ρ = −0.33" and "None".

SAY THIS

The last part asks why the numbers don't line up. We took public data on every county and tested which characteristics go with recording more or less than expected.

We expected hardship to matter: poverty, households without a car, SNAP participation, how rural a county is. None of them predicted the gaps.

What did predict them is access. The strongest single factor is the share of residents whose nearest food shelf is in another county. The more residents are closer to a food shelf across the county line, the less their own county records. Next come the share of residents living more than 10 miles from a food shelf, and how many food shelves a county has per resident.

The cross-county factor also held when we tested everything together in one model, holding the other factors constant.

What it means

We tested each characteristic three ways:

1. **Group comparison:** do lower-than-expected counties differ from counties with no mismatch?
2. **Correlation:** across all 86 counties, does it rise and fall with the size of the gap?
3. **Regression:** does it still matter when the other factors are held constant?

Why poverty doesn't show up: Feeding America's estimate already uses poverty and unemployment to estimate need, so poverty is already built into "expected". What's left over is about access, not hardship.

Why rural share doesn't show up: fully rural counties are on **both** lists. Roseau, Norman and Lac qui Parle are low; Cook, Kittson, Mahnomen and Renville are high. Being rural isn't the barrier; distance within rural areas is.

Numbers to know

Characteristic	Correlation with the gap	Significant?
Nearest food shelf in another county	−0.33 (p = 0.002)	Yes
Residents more than 10 miles away	−0.27	Yes
Food shelves per resident	+0.25	Yes
Poverty rate	−0.16	No
No-vehicle households	−0.09	No
SNAP participation	−0.06	No
Rural share	+0.06	No

- Negative = more of it goes with recording **less** than expected.

- Regression: the cross-county factor is the only clear predictor (−0.26 per standard deviation, p = 0.008). The model explains 22% of the differences.
- Typical share of residents whose nearest food shelf is in another county: 20% (lower), 12% (no mismatch), 5% (higher).

If a judge asks

What does p = 0.002 mean?

If there were really no relationship, a result this strong would show up by chance only about 2 times in 1,000.

Why doesn't car access matter?

Most counties sit around 5% of households without a car, and the Census margin of error is about as big as the differences. Carless households are mostly in towns, close to food shelves. Mahnomen has the highest no-car rate of the focus counties and still records more than expected.

22% explained, isn't that low?

For county-level social data it's reasonable, and we say openly that most of the variation is unexplained, for example local volunteer capacity, opening hours, or how each site counts visits.

Slide 6: Activity is counted where the food shelf is LIVE MAP

On the slide: the map zoomed on Nicollet with a 49% arrow into Blue Earth, a "49%" card, "Nicollet 0.29× → Blue Earth 1.89×", and a Wilkin card "6 of 9".

SAY THIS

Why would access matter so much? Because food shelf activity is counted where the food shelf is, not where the people it serves live. If your nearest food shelf is across the county line, your visit counts for the other county.

[Alt+Tab to the map. Press 2 for Cross-county. Click NICOLLET.]

Nicollet is the clearest example. For about half of Nicollet's residents, the nearest food shelf is in Blue Earth County, around Mankato. Nicollet records only 0.29 times what's expected. Blue Earth, right next door, records almost twice what's expected, from a few very busy food shelves. A quiet county right next to a busy one is exactly what you'd see if people cross the line.

[Click WILKIN.]

Wilkin is the extreme case. It has no food shelf at all in the data, so every resident's nearest food shelf is in another county, about 19 miles away on average. Yet The Food Group's helpline logged 9 calls from Wilkin, and referred 6 of them to a food shelf. Those referrals have to be to food shelves in neighboring counties. So we know cross-county use happens.

To be clear, Nicollet is our one clear case. We don't claim this explains every county; that's why collecting client ZIP codes is our first recommendation.

[Alt+Tab back to the slides and move to slide 7.]

What it means

The percentages come from Census data, not from guessing:

1. The Census splits Minnesota into 4,706 neighborhoods (block groups of about 1,000–3,000 people) and gives each one's population and center point.
2. We have the location of each of the 435 fixed food shelves.
3. For each neighborhood, we found the nearest food shelf by straight-line distance and noted which county it's in.
4. Adding up the people gives each county's share. For Nicollet: 41% Nicollet, 49% Blue Earth, 10% Brown.

"Nearest" isn't the same as "where people go". People may use a food shelf near work or family. Only client ZIP codes would show real trips.

Numbers to know

County	Records vs. expected	Residents whose nearest food shelf is in another county
Nicollet	0.29×	59% (49% Blue Earth, 10% Brown)
Blue Earth (neighbor)	1.89×	not shortlisted
Wilkin	no food shelf	100% (70% Otter Tail, 30% Clay)

- Wilkin: about 6,300 people, an estimated 720–810 food insecure, 0 food shelves, 9 helpline calls, 6 referred to a food shelf.

If a judge asks

Is cross-county use just an assumption?

It's inferred, not observed, because we don't have client addresses. But three things point the same way: it's the strongest statistical factor and holds in the regression; Nicollet sits next to a busy Blue Earth; and Wilkin's residents are referred to food shelves even though Wilkin has none.

What about other counties, like Pope?

Pope's neighbors are also below expected, so cross-county use doesn't explain Pope. For Pope it's few, quiet food shelves and distance. That's why we say "some counties, not all".

Why don't Norman residents go to Mahnomen, which is high?

For 92% of Norman residents the nearest food shelf is in Norman, and Mahnomen's food shelves are about 25 miles away on average. Some may still go, but we can't see that without ZIP codes.

Wilkin's neighbors aren't busy. Doesn't that contradict it?

Wilkin has only about 6,300 people, too few to visibly move a county the size of Otter Tail (about 60,000).

Slide 7: Distance matters; mobile pantries may help LIVE MAP

On the slide: the distance map and four cards: "25% vs 9%", "0 of 10", "2.4×" (Cook) and "1.23×" (far-away counties with a mobile pantry), with a note that the mobile pantry result is suggestive.

SAY THIS

Distance matters too, even when the nearest food shelf is in your own county.

[Alt+Tab to the map. Press 3 for Distance.]

Each dot here is a neighborhood, and the darker it is, the farther people live from a fixed food shelf. Most Minnesotans, about 84%, live within 5 miles of one. But in the counties that record less than expected, typically a quarter of residents live more than 10 miles away, compared with about 9% in other counties.

Lac qui Parle, Norman and Roseau, three of our five lower counties, all have 30 to 38% of residents more than 10 miles from a food shelf. For Norman and Roseau, over 90% of residents' nearest food shelf is in their own county, so the issue there is distance, not crossing a county line.

[Tick "Outline counties with a mobile pantry".]

A mobile pantry is a truck that brings food to a town on a schedule. The green outlines are counties that have one. None of the 10 lower-than-expected counties has a mobile pantry.

[Click COOK.]

Cook County is the clearest example. It's very remote: 48% of residents live more than 10 miles from a fixed food shelf. But 64% of its visits come from mobile pantries, and it records 2.4 times what's expected. More broadly, among counties where many people live far away, the ones with a mobile pantry record about 1.23 times expected, and the ones without about 0.8 times.

With so few counties this is suggestive, not proven, but it points to a concrete action.

[Alt+Tab back to the slides.]

What it means

Distance is measured the same way as on slide 6: from each neighborhood's population center to the nearest fixed food shelf, as a straight line, weighted by population so a county's figure reflects where its people actually live.

Mobile pantries come from the data: 61 of the 93 "Outreach Programs" in Organization.xlsx are mobile. 21 of the 86 scored counties have one.

The mobile pantry result is "suggestive" for three reasons: the groups are small (only 7 far-away counties have one), mobile pantry visits count toward a county's activity (so the link is partly built in), and they're recorded under the county they're based in.

Numbers to know

Measure	Lower counties	Other counties
Typical share of residents > 10 miles away	25%	9%
Counties with a mobile pantry	0 of 10	28% (no mismatch), 25% (higher)

Far-away counties (43)	Activity vs. expected
With a mobile pantry (7)	1.23×
Without (36)	0.80×

- Lower vs. no-mismatch, distance: p ≈ 0.04. Mobile pantry yes/no: p ≈ 0.10. Far-away comparison: p ≈ 0.05.
- Cook: 48% of residents > 10 miles away, 64% of visits from mobile pantries, 2.4× expected.

If a judge asks

Why straight-line distance?
It's simple and consistent. Real driving distance is longer, especially around lakes and in the north, so our numbers understate the problem. Driving distance is a next step.

Do mobile pantries cause higher activity?
We can't say that. It's an association with few counties. That's why we present it as a recommendation to try, not a proven effect.

Why do many dots near the Twin Cities look light?
They're close to food shelves. The dark dots, the far-away neighborhoods, are mostly in the west and north.

Slide 8: The supply network and site size matter too

On the slide: a bar chart of the typical county for each main food bank (0.82×, 0.91×, 1.12×, 1.30×), plus cards "0.22 → 0.34" and "~1,750".

SAY THIS

Two more things help explain the gaps.

The first is the food bank network. Food shelves get their food from a regional food bank. When we grouped counties by the food bank that supplies most of their visits, a clear pattern showed up. Counties supplied by North Country Food Bank, in the northwest, typically record less than expected, about 0.82 times. Counties supplied by Channel One, around Rochester, and Second Harvest Northland, in the northeast, typically record more, and not one of their counties is below expected.

Adding the food bank to our model raises how much of the gaps we can explain from 22% to 34%, and the cross-county factor still holds. So these are two separate effects. Because each food bank serves its own region, we can't tell whether this comes from the food bank's practices, like how much food it sends or how its partners record visits, or from something else about the region. So it's a lead to explore with The Food Group, not a verdict on any food bank.

The second is site size. The biggest "higher" counties, Washington, Steele, Rice and Olmsted, have a few very busy food shelves close to where people live. Washington averages about 1,750 visits per site per month, the busiest in the state. And in most of those counties about half of food-insecure residents earn too much to qualify for SNAP, so food shelves may be their main help.

What it means

The food bank test is a Kruskal-Wallis test, which asks whether counties in different food bank groups really differ or whether the gap could be chance. The answer is p = 0.02, so it's probably real. Three of the five shortlisted lower counties, Norman, Roseau and Pope, are North Country counties.

The SNAP point is a **supporting pattern**, not a main finding. Higher-than-expected counties have more food-insecure people above the SNAP income limit (37% vs. 31%), but it isn't significant (p ≈ 0.15). It fits Washington (55%), Olmsted (48%), Dodge (46%) and Rice (45%), but not Steele (35%, near the state median of 33%).

Numbers to know

Main food bank	Counties	Typical county	Lower / higher counties
North Country	21	0.82×	3 / 2
Second Harvest Heartland	39	0.91×	7 / 5
Channel One	13	1.12×	0 / 5
Second Harvest Northland	11	1.30×	0 / 3

- Model: R^2 0.22 without the food bank, 0.34 with it. North Country vs. Channel One: −0.66, p = 0.02. Cross-county factor stays significant (p = 0.006).
- Visits per site per month: Washington ~1,750, Steele ~1,160, Olmsted ~920, Rice ~890. State median ~177.

If a judge asks

Is North Country doing a worse job?

No. Each food bank covers one region, so the difference may come from the region, such as geography and how spread out people are. It's a question to ask, not a judgment.

Second Harvest Heartland has the most lower counties. Why isn't it lowest?

It serves the most counties (39), so it has more of everything. Its typical county is close to expected (0.91×).

Are the busy hubs busy because they serve neighbors?

Our nearest-food-shelf data doesn't show many outsiders closer to them, so we don't claim that. Their story is very busy sites close to residents, with many food-insecure people who can't get SNAP.

What's R^2?

The share of the differences between counties that the model explains: 0.34 means about a third.

Slide 9: Takeaway

On the slide: "Where food shelves are matters as much as how much need there is" and three numbered recommendations.

SAY THIS

So, to answer Question 1. Feeding America's estimates and The Food Group's numbers agree on county size and on the direction over time, but not per resident. Pounds tracks need best, by a small margin. Twelve counties break the pattern. And what explains it isn't hardship. It's access: where food shelves are, how big they are, and how far people have to travel.

Because activity is counted where the food shelf is, a county can look under-served when its residents are actually being served next door.

That leads to three recommendations.

First, collect client ZIP codes, even for a sample of visits. That's the only way to see cross-county use directly and confirm what we can only infer today.

Second, send mobile pantries to the counties with the biggest distance gaps: Lac qui Parle, Norman and Roseau. A third or more of their residents live more than 10 miles from a food shelf, and none has a mobile pantry.

Third, compare food banks: how much food each one supplies, and how its partner sites record visits, especially in the North Country network.

What it means

The takeaway sentence is the one line judges should remember. It ties all four parts together: the need estimate alone is a weak guide, and location shapes the recorded numbers. Each recommendation comes straight from a finding:

Recommendation	Comes from
Collect client ZIP codes	Cross-county use is our strongest factor but can't be observed without client addresses (slide 6)
Mobile pantries for distance gaps	Lower counties are farther from food shelves; none has a mobile pantry; Cook shows what one can do (slide 7)
Compare food banks	The network effect is significant but regional, so it needs a closer look (slide 8)

Numbers to know

- Five lower counties: Nicollet, Pope, Lac qui Parle, Norman, Roseau. Seven higher: Steele, Washington, Rice, Renville, Mahnomen, Pipestone, Olmsted.

If a judge asks

What would you do with more time?

Use driving distance and mobile pantry routes instead of straight lines, add 2025 when Map the Meal Gap releases it, and re-check two one-year cases: Scott (low in 2024) and Wabasha (high in 2024).

What's the biggest limitation?

We can't see where clients live, so cross-county use is inferred. Also, the need numbers are model estimates, and we have only three years.

How should The Food Group use this?

Plan with per-resident, site-level and travel-distance views, not county totals alone.

Demo do's and don'ts

The live map is the part judges will remember, so script every click and avoid the spots that invite a hard question.

Do

- Rehearse the clicks: tab 1 slider to 3× and back; press 2, click Nicollet, then Wilkin; press 3, tick the mobile pantry box, click Cook.
- Use "← Statewide view" on tab 2 if you need to zoom back out.
- Hover any county on tab 3 to read its numbers if a judge asks about it.
- If the map won't open or freezes, stay on the slide. Each live slide shows the same result as a picture.

Don't

- Don't click **Pope**, **Lac qui Parle** or the busy hubs (**Steele, Washington, Rice**) on the cross-county tab. Their neighbors don't support the cross-county story.
- Don't explore unplanned on stage. It eats time and looks messy.

Wording traps

Avoid saying	Say instead	Why
Nicollet residents use Blue Earth's food shelves	Half of Nicollet's residents live closer to a food shelf in Blue Earth	We know distance, not where people actually go
These counties are under-served	These counties record less than expected	Their residents may be served next door
Mobile pantries increase activity	Counties with mobile pantries record more; it's suggestive	Few counties, partly built into the numbers
North Country performs worse	North Country counties record less; it may be regional	Food bank and region can't be separated
Busy hubs serve their neighbors	Busy hubs have a few very busy sites close to residents	The data doesn't show many outsiders closer to them
Poverty doesn't matter	Poverty doesn't explain the gaps	It's already built into the need estimate

Likely judge questions (general)

These cover Q1 as a whole; each slide section has its own questions too.

Why Spearman and not just Pearson?

Minnesota has one huge county (Hennepin) and many tiny ones. Pearson uses actual values, so one extreme county can pull it. Spearman uses rankings, so it can't. We report both, and where they disagree we checked which counties drive the result.

Isn't the cross-county factor just geography?

It holds even after controlling for distance, rural share, poverty and site density in the regression, and it stays significant when the food bank network is added. So it isn't standing in for those.

Did you test anything else?

Yes: SuperShelf certification and how many people commute to another county for work. Neither showed any link. The commuting result rules out "people use a food shelf near their job": the pattern is about where the nearest food shelf is.

Why is Mahnomen higher than expected?

It has the highest poverty of the focus counties (19.9%) and the highest no-car rate (8.7%), but many food shelves close to residents. Very high hardship plus good access. The need estimate may understate need there, but that's a hypothesis. Mahnomen is also mostly within the White Earth Reservation.

Why is Steele four times expected?

Owatonna is a regional center with a few very busy food shelves, short distances, and activity growing every year. Poverty is below the state median, so it isn't hardship.

Renville is high but far from food shelves. Doesn't that break your story?

Yes, and we say so. Renville has busy sites despite 45% of residents living more than 10 miles away. It's a counterexample, which is why we say access explains some of the gaps, not all.

Why is Wilkin left out?

It has no food shelf in the data, so there's nothing to compare. We kept it in our tables, and it's useful evidence that cross-county use happens.

How much of the mismatch do you explain?

About a fifth with county characteristics (R^2 0.22), and about a third once the food bank network is added (0.34). The rest is likely local factors we don't have data on.

What data would change your conclusions?

Client ZIP codes. If they showed Nicollet residents don't actually go to Blue Earth, the cross-county explanation would weaken. If they showed they do, it would become direct evidence.